

EXP1 > Experiment1_DFA.c > ...

```
1  ∨ #include <stdio.h>
2  #include <string.h>
3  ∨ int main()
4  {
5      char str[100];
6      int state = 0;
7      int i;
8      printf("Enter the string: ");
9      scanf("%s", str);
10 ∨ for(i = 0; str[i] != '\0'; i++)
11 {
12 ∨     switch(state)
13     {
14         case 0:
15             if(str[i] == 'a')
16                 state = 1;
17             else
18                 state = -1;
19             break;
20
21         case 1:
22             if(str[i] == 'b')
23                 state = 2;
24             else
25                 state = -1;
26             break;
27
28         case 2:
29             state = -1;
30             break;
31     }
32
33     if(state == -1)
34         break;
35 }
36 if(state == 2 && str[i] == '\0')
37     printf("\nString Accepted.");
38 else
39     printf("\nString Rejected.");
40
41 return 0;
42 }
```

- PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_DFA.exe
Enter the string: ab

String Accepted.
- PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_DFA.exe
Enter the string: abb

String Rejected.
- PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_DFA.exe
Enter the string: ba

EXP1 > Experiment1_Assignment1.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      char str[100];
5      int state = 0;
6      int i;
7      printf("Enter the string: ");
8      scanf("%s", str);
9      for(i = 0; str[i] != '\0'; i++)
10     {
11         switch(state)
12         {
13             case 0:
14                 if(str[i] == 'a')
15                     state = 1;
16                 else
17                     state = -1;
18                 break;
19             case 1:
20                 if(str[i] == 'b')
21                     state = 2;
22                 else
23                     state = -1;
24                 break;
25             case 2:
26                 if(str[i] == 'b')
27                     state = 3;
28                 else
29                     state = -1;
30                 break;
31             case 3:
32                 state = -1;
33                 break;
34         }
35         if(state == -1)
36             break;
37     }
38     if(state == 3 && str[i] == '\0')
39         printf("\nString Accepted.");
40     else
41         printf("\nString Rejected.");
42     return 0;
43 }
```

● PS C:\Users\Afroz\Desktop\PCD\EXP1> gcc Experiment1_Assignment1.c -o Experiment1_Assignment1

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment1.exe

Enter the string: abb

String Accepted.

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment1.exe

Enter the string: ab

String Rejected.

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment1.exe

Enter the string: abba

String Rejected.

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment1.exe

Enter the string: abc

String Rejected.

EXP1 > Experiment1_Assignment2.c > ...

```
1  #include <stdio.h>
2  int main()
3  {
4      char str[100];
5      int state = 0;
6      int i;
7      printf("Enter the string: ");
8      scanf("%s", str);
9      for(i = 0; str[i] != '\0'; i++)
10     {
11         switch(state)
12         {
13             case 0:
14                 if(str[i] == 'a')
15                     state = 1;
16                 else
17                     state = -1;
18                 break;
19             case 1:
20                 if(str[i] == 'b')
21                     state = 2;
22                 else
23                     state = -1;
24                 break;
25             case 2:
26                 if(str[i] == 'c')
27                     state = 3;
28                 else
29                     state = -1;
30                 break;
31             case 3:
32                 state = -1;
33                 break;
34         }
35         if(state == -1)
36             break;
37     }
38     if(state == 3 && str[i] == '\0')
39         printf("\nString Accepted.");
40     else
41         printf("\nString Rejected.");
42     return 0;
43 }
```

● PS C:\Users\Afroz\Desktop\PCD\EXP1> gcc Experiment1_Assignment2.c -o Experiment1_Assignment2

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment2.exe

Enter the string: abc

String Accepted.

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment2.exe

Enter the string: abb

String Rejected.

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment2.exe

Enter the string: aab

String Rejected.

EXP1 > C Experiment1_Assignment3.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      char str[100];
5      int state = 0;
6      int i;
7      printf("Enter the string: ");
8      scanf("%s", str);
9      for(i = 0; str[i] != '\0'; i++)
10     {
11         switch(state)
12         {
13             case 0:
14                 if(str[i] == 'a')
15                     state = 1;
16                 else
17                     state = -1;
18                 break;
19             case 1:
20                 if(str[i] == 'a')
21                     state = 2;
22                 else
23                     state = -1;
24                 break;
25             case 2:
26                 state = -1;
27                 break;
28         }
29         if(state == -1)
30             break;
31     }
32     if(state == 2 && str[i] == '\0')
33         printf("\nString Accepted.");
34     else
35         printf("\nString Rejected.");
36     return 0;
37 }
```

● PS C:\Users\Afroz\Desktop\PCD\EXP1> gcc Experiment1_Assignment3.c -o Experiment1_Assignment3

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment3.exe

Enter the string: aa

String Accepted.

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment3.exe

Enter the string: ab

String Rejected.

● PS C:\Users\Afroz\Desktop\PCD\EXP1> .\Experiment1_Assignment3.exe

Enter the string: ba

String Rejected.